C h a p t e r 3

Half-Wave Rectifiers

The Basics of Analysis

3.1 INTRODUCTION

A rectifier converts AC to DC. The purpose of a rectifier may be to produce an output that is purely DC, or the purpose may be to produce a voltage or current waveform that has a specified DC component.

In practice, the half-wave rectifier is used most often in low-power applications because the average current in the supply will not be zero, and nonzero average current may cause problems in transformer performance. While practical applications of this circuit are limited, it is very worthwhile to analyze the half-wave rectifier in detail. Athorough understanding of the half-wave rectifier circuit will enable the student to advance to the analysis of more complicated circuits with a minimum of effort.

The objectives of this chapter are to introduce general analysis techniques for power electronics circuits, to apply the power computation concepts of the previous chapter.

3.2 Resistive Load

Creating a DC Component Using an Electronic Switch

A basic half-wave rectifier with a resistive load is shown in Fig. 3-1*a*. The source is AC, and the objective is to create a load voltage that has a nonzero DC component.

The diode is a basic electronic switch that allows current in one direction only. For the positive half-cycle of the source in this circuit, the diode is on (forward-biased).

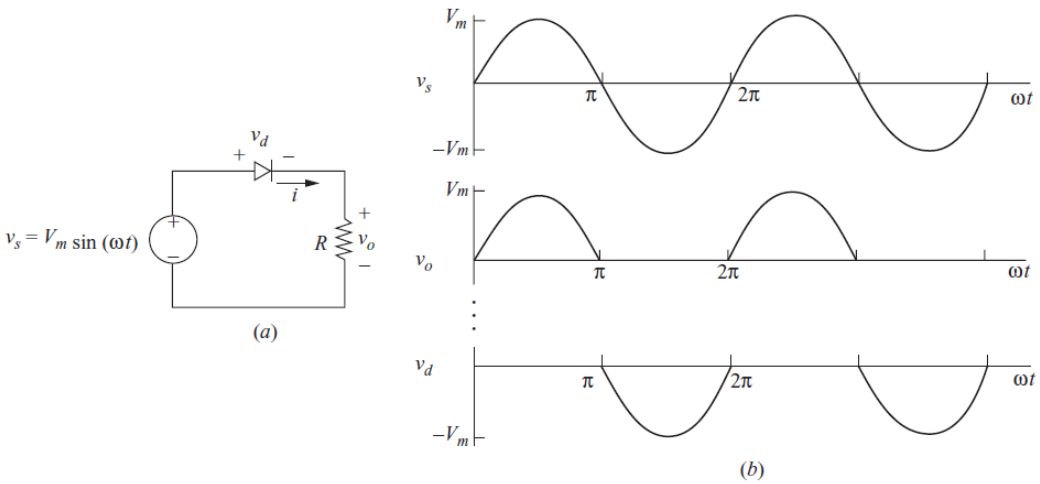


Figure 3-1 (*a*) Half-wave rectifier with resistive load; (*b*) Voltage waveforms.

Considering the diode to be ideal, the voltage across a forward-biased diode is zero and the current is positive.

For the negative half-cycle of the source, the diode is reverse-biased, making the current zero. The voltage across the reverse-biased diode is the source voltage, which has a negative value.

The voltage waveforms across the source, load, and diode are shown in Fig. 3-1*b*. Note that the units on the horizontal axis are in terms of angle (ωt). This representation is useful because the values are independent of frequency.

The DC component V_O of the output voltage is the average value of a half-wave rectified sinusoid

$$V_o = V_{\text{avg}} = \frac{1}{2\pi}\int_0^{\pi} V_m \sin(\omega t)\, d(\omega t) = \frac{V_m}{\pi} \qquad (3\text{-}1)$$

The DC component of the current for the purely resistive load is

$$I_o = \frac{V_o}{R} = \frac{V_m}{\pi R} \qquad (3\text{-}2)$$

Average power absorbed by the resistor in Fig. 3-1a can be computed from $P = I_{rms}^2 R = V_{rms}^2/R$. When the voltage and current are half-wave rectified sine waves,

$$V_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{\pi}[V_m \sin(\omega t)]^2 d(\omega t)} = \frac{V_m}{2}$$

$$I_{\text{rms}} = \frac{V_m}{2R} \qquad (3\text{-}3)$$

In the preceding discussion, the diode was assumed to be ideal. For a real diode, the diode voltage drop will cause the load voltage and current to be reduced, but not appreciably if V_m is large. For circuits that have voltages much larger than the typical diode drop, the improved diode model may have only second-order effects on the load voltage and current computations.

Example 3-1(Half-Wave Rectifier with Resistive Load)
For the half-wave rectifier of Fig. 3-1a, the source is a sinusoid of $120\ V\ rms$ at a frequency of 60 Hz. The load resistor is 5 Ω. Determine (a) the average load current, (b) the average power absorbed by the load and (c) the power factor of the circuit.
■ Solution
(a) The voltage across the resistor is a half-wave rectified sine wave with peak value $Vm = 120\sqrt{2} = 169.7\ V$. From Eq. (3-2), the average voltage is $\frac{V_m}{\pi}$, and the average current is:

$$I_o = \frac{V_o}{R} = \frac{V_m}{\pi R} = \frac{\sqrt{2}(120)}{5\pi} = 10.8\ \text{A}$$

(b) From Eq. (3-3), the rms voltage across the resistor for a half-wave rectified sinusoid is

$$V_{\text{rms}} = \frac{V_m}{2} = \frac{\sqrt{2}(120)}{2} = 84.9\ \text{V}$$

The power absorbed by the resistor is

$$P = \frac{V_{\text{rms}}^2}{R} = \frac{84.9^2}{4} = 1440\ \text{W}$$

The rms current in the resistor is $V_m/(2R) = 17.0\ A$, and the power could also be calculated from $I_{rms}^2\ R = (17.0)^2 \times (5) = 1440\ W$.
(c) The power factor is:

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{s,\text{rms}} I_{s,\text{rms}}} = \frac{1440}{(120)(17)} = 0.707$$

3.3 Resistive-Inductive Load

Industrial loads typically contain inductance as well as resistance. As the source voltage goes through zero, becoming positive in the circuit of Fig. 3 2a, the diode becomes forward-biased. The Kirchhoff voltage law equation that describes the current in the circuit for the forward-biased ideal diode is:

$$V_m \sin(\omega t) = Ri(t) + L\frac{di(t)}{dt} \qquad (3\text{-}4)$$

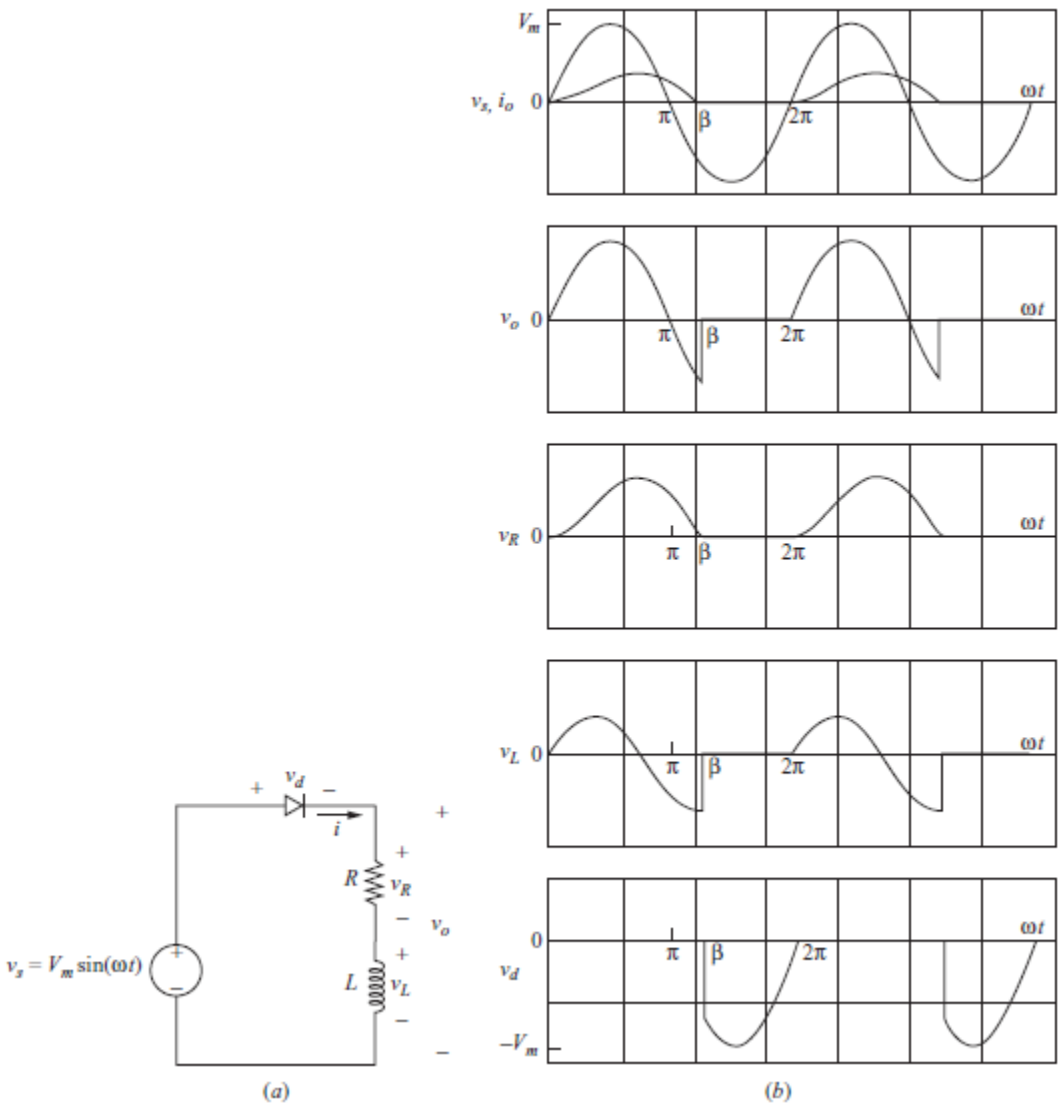


Figure 3-2 (*a*) Half-wave rectifier with an *RL* load; (*b*) Waveforms.

The solution can be obtained by expressing the current as the sum of the forced response and the natural response:

$$i(t) = i_f(t) + i_n(t) \qquad (3\text{-}5)$$

The forced response for this circuit is the current that exists after the natural response has decayed to zero. In this case, the forced response is the steady-state sinusoidal current that would exist in the circuit if the diode were not present.

This steady-state current can be found from phasor analysis, resulting in:

$$i_f(t) = \frac{V_m}{Z}\sin(\omega t - \theta) \qquad (3\text{-}6)$$

where $Z = \sqrt{R^2 + (\omega L)^2}$ and $\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$

The natural response is the transient that occurs when the load is energized. It is the solution to the homogeneous differential equation for the circuit without the source or diode:

$$R\,i(t) + L\frac{di(t)}{dt} = 0 \qquad (3\text{-}7)$$

For this first-order circuit, the natural response has the form

$$i_n(t) = Ae^{-t/\tau} \qquad (3\text{-}8)$$

where τ is the time constant $\tau = L/R$ and A is a constant that is determined from the initial condition. Adding the forced and natural responses gets the complete solution.

$$i(t) = i_f(t) + i_n(t) = \frac{V_m}{Z}\sin(\omega t - \theta) + Ae^{-t/\tau} \qquad (3\text{-}9)$$

The constant A is evaluated by using the initial condition for current. The initial condition of current in the inductor is zero because it was zero before the diode started conducting and it cannot change instantaneously.

Using the initial condition and Eq. (3-9) to evaluate A yields:

$$i(0) = \frac{V_m}{Z}\sin(0 - \theta) + Ae^0 = 0$$

$$A = -\frac{V_m}{Z}\sin(-\theta) = \frac{V_m}{Z}\sin\theta \qquad (3\text{-}10)$$

Substituting for A in Eq. (3-9) gives

$$i(t) = \frac{V_m}{Z}\sin(\omega t - \theta) + \frac{V_m}{Z}\sin(\theta)e^{-t/\tau}$$

$$= \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-t/\tau}\right] \qquad (3\text{-}11)$$

It is often convenient to write the function in terms of the angle (ωt)rather than time. This merely requires(ωt) to be the variable instead of (t). To write the above equation in terms of angle, t in the exponential must be written as (ωt), which requires (τ) to be multiplied by (ω) also. The result is:

$$i(\omega t) = \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-\omega t/\omega\tau}\right] \qquad (3\text{-}12)$$

A typical graph of circuit current is shown in Fig. 3-2b. Equation (3-12) is valid for positive currents only because of the diode in the circuit, so current is zero when the function in Eq. (3-12) is negative. When the source voltage again becomes positive, the diode turns on, and the positive part of the waveform in Fig. 3-2b is repeated. This occurs at every positive half-cycle of the source. The voltage waveforms for each element are shown in Fig. 3-2b.

Note that the diode remains forward-biased longer than π rad and that the source is negative for the last part of the conduction interval. This may seem unusual, but an examination of the voltages reveals that Kirchhoff's voltage law is satisfied and there is no contradiction. Also note that the inductor voltage is negative when the current is decreasing $(vL = L\ di/dt)$.

The point when the current reaches zero in Eq. (3-12) occurs when the diode turns off. The first positive value of (ωt)in Eq. (3-12) that results in zero current is called the extinction angle β.

Substituting $\omega t = \beta$in Eq. (3-12), the equation that must be solved is:

$$i(\beta) = \frac{V_m}{Z}\left[\sin(\beta - \theta) + \sin(\theta)e^{-\beta/\omega\tau}\right] = 0 \qquad (3\text{-}13)$$

Which reduces to:

$$\sin(\beta - \theta) + \sin(\theta)e^{-\beta/\omega\tau} = 0 \qquad (3\text{-}14)$$

There is no closed-form solution for β, and some numerical method is required. To summarize, the current in the half-wave rectifier circuit with RL load (Fig. 3-2) is expressed as:

$$i(\omega t) = \begin{cases} \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-\omega t/\omega\tau}\right] & \text{for } 0 \le \omega t \le \beta \\ 0 & \text{for } \beta \le \omega t \le 2\pi \end{cases} \qquad (3\text{-}15)$$

where $Z = \sqrt{R^2 + (\omega L)^2}$ $\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$ and $\tau = \frac{L}{R}$

The average power absorbed by the load is $I_{rms}^2 R$, since the average power absorbed by the inductor is zero. The rms value of the current is determined from the current function of Eq. (3-15).

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} i^2(\omega t)\, d(\omega t)} = \sqrt{\frac{1}{2\pi}\int_0^{\beta} i^2(\omega t)\, d(\omega t)} \qquad (3\text{-}16)$$

Average current is:

$$I_o = \frac{1}{2\pi}\int_0^{\beta} i(\omega t)\, d(\omega t) \qquad (3\text{-}17)$$

Example 3-2 (Half-Wave Rectifier with RL Load)

For the half-wave rectifier of Fig. 3-2a, $= 100\ \Omega, L = 0.1\ H, \omega t = 377\ rad/s, and\ V_m = 100\ V$. Determine (a) an expression for the current in this circuit, (b) the average current, (c) the rms current, (d) the power absorbed by the RL load, and (e) the power factor.

■ **Solution**

For the parameters given,

$$Z = [R^2 + (\omega L)^2]^{0.5} = 106.9\,\Omega$$
$$\theta = \tan^{-1}(\omega L/R) = 20.7° = 0.361 \text{ rad}$$
$$\omega t = \omega L/R = 0.377 \text{ rad}$$

(a) Equation (3-15) for current becomes

$$i(\omega t) = 0.936\sin(\omega t - 0.361) + 0.331e^{-\omega t/0.377} \quad \text{A} \quad \text{for } 0 \le \omega t \le \beta$$

Beta is found from Eq. (3-14).

$$\sin(\beta - 0.361) + \sin(0.361)e^{-\beta/0.377} = 0$$

Using a numerical root-finding program, β is found to be 3.50 rad, or 201°.

(b) Average current is determined from Eq. (3-17).

$$I_o = \frac{1}{2\pi}\int_0^{3.50} \left[0.936\sin(\omega t - 0.361) + 0.331e^{-\omega t/0.377}\right] d(\omega t) = 0.308 \text{ A}$$

(A numerical integration program is recommended.)

(c) The rms current is found from Eq. (3-16) to be

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{3.50} \left[0.936\sin(\omega t - 0.361) + 0.331e^{-\omega t/0.377}\right]^2 d(\omega t)} = 0.474 \text{ A}$$

(d) The power absorbed by the resistor is:

$$P = I_{\text{rms}}^2 R = (0.474)^2(100) = 22.4 \text{ W}$$

The average power absorbed by the inductor is zero. Also P can be computed from the definition of average power:

$$P = \frac{1}{2\pi}\int_0^{2\pi} p(\omega t)\,d(\omega t) = \frac{1}{2\pi}\int_0^{2\pi} v(\omega t)i(\omega t)\,d(\omega t)$$

$$= \frac{1}{2\pi}\int_0^{3.50} [100\sin(\omega t)]\left[0.936\sin(\omega t - 0.361) + 0.331e^{-\omega t/0.377}\right] d(\omega t)$$

$$= 22.4 \text{ W}$$

(e) The power factor is computed from the definition $pf = P/S$, and P is the power supplied by the source, which must be the same as that absorbed by the load.

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{s,\text{rms}} I_{\text{rms}}} = \frac{22.4}{\left(100/\sqrt{2}\right)0.474} = 0.67$$

Note that the power factor is not $cos\ \theta$.

3.5 *RL* and DC Source Load(Supplying Power to a DC Source from an AC Source)

Another variation of the half-wave rectifier is shown in Fig. 3-5*a*. The load consists of a resistance, an inductance, and a DC voltage.

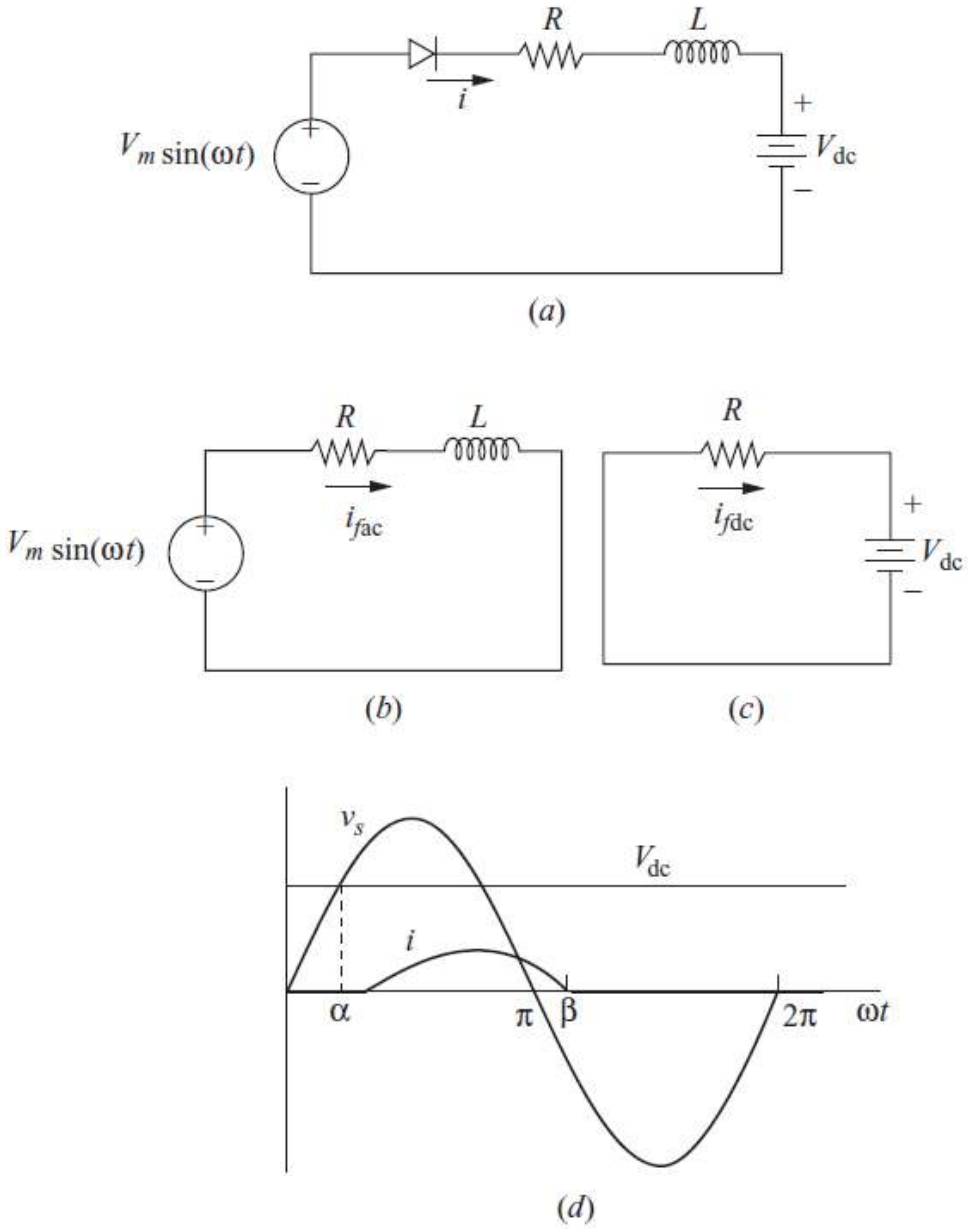


Figure 3-5 (*a*) Half-wave rectifier with *RL DC* source load; (*b*) Circuit for forced response from ac source; (*c*) Circuit for forced response from DC source; (*d*) Waveforms.

Starting the analysis at $(\omega t = 0)$ and assuming the initial current is zero, recognize that the diode will remain off as long as the voltage of the AC source is less than the DC voltage.

Letting α be the value of ωtthat causes the source voltage to be equal to V_{dc},

$$V_m \sin \alpha = V_{dc}$$

Or

$$\alpha = \sin^{-1}\left(\frac{V_{dc}}{V_m}\right) \tag{3-18}$$

The diode starts to conduct at $\omega t = \alpha$. With the diode conducting, Kirchhoff's voltage law for the circuit yields the equation:

$$V_m \sin(\omega t) = Ri(t) + L\frac{di(t)}{dt} + V_{dc} \tag{3-19}$$

Total current is determined by summing the forced and natural responses:

$$i(t) = i_f(t) + i_n(t)$$

The current $if\ (t)$ is determined using superposition for the two sources. The forced response from the ac source (Fig. 3-5b) is $(V_m/Z)\ sin(\omega t - \theta)$. The forced response due to the dc source (Fig. 3-5c) is $(-V_{dc}/R)$. The entire forced response is:

$$i_f(t) = \frac{V_m}{Z}\sin(\omega t - \theta) - \frac{V_{dc}}{R} \tag{3-20}$$

The natural response is

$$i_n(t) = Ae^{-t/\tau} \tag{3-21}$$

Adding the forced and natural responses gives the complete response.

$$i(\omega t) = \begin{cases} \frac{V_m}{Z}\sin(\omega t - \theta) - \frac{V_{dc}}{R} + Ae^{-\omega t/\omega\tau} & \text{for } \alpha \le \omega t \le \beta \\ 0 & \text{otherwise} \end{cases} \tag{3-22}$$

The extinction angle β is defined as the angle at which the current reaches zero, as was done earlier in Eq. (3-15). Using the initial condition of $i(\alpha) = 0$ and solving for A,

$$A = \left[-\frac{V_m}{Z} \sin(\alpha - \beta) + \frac{V_{dc}}{R} \right] e^{\alpha/\omega\tau} \qquad (3\text{-}23)$$

Figure 3-5d shows voltage and current waveforms for a half-wave rectifier with RL-source load.

The average power absorbed by the resistor is $I_{rms}^2 \times R$, where

$$I_{rms} = \sqrt{\frac{1}{2\pi} \int_{\alpha}^{\beta} i^2(\omega t)\, d(\omega t)} \qquad (3\text{-}24)$$

The average power absorbed by the DC source is:

$$P_{dc} = I_o V_{dc} \qquad (3\text{-}25)$$

Where I_o is the average current, that is,

$$I_o = \frac{1}{2\pi} \int_{\alpha}^{\beta} i(\omega t)\, d(\omega t) \qquad (3\text{-}26)$$

Assuming the diode and the inductor to be ideal, there is no average power absorbed by either. The power supplied by the AC source is equal to the sum of the power absorbed by the resistor and the DC source:

$$P_{ac} = I_{rms}^2 R + I_o V_{dc} \qquad (3\text{-}27)$$

Or it can be computed from:

$$P_{ac} = \frac{1}{2\pi} \int_{0}^{2\pi} v(\omega t)\, i(\omega t)\, d(\omega t) = \frac{1}{2\pi} \int_{\alpha}^{\beta} (V_m \sin \omega t)\, i(\omega t)\, d(\omega t) \qquad (3\text{-}28)$$

Example 3-5 (Half-Wave Rectifier with *RL*-Source Load)

For the circuit of Fig. 3-5*a*, $R = 2\ \Omega, L = 20\ mH$, and $V_{dc} = 100\ V$. The AC source is 120 V rms at 60 Hz. Determine (*a*) an expression for the current in the circuit, (*b*) the power absorbed by the resistor, (*c*) the power absorbed by the DC source, and (*d*) the power supplied by the AC source and the power factor of the circuit.

■ **Solution**

From the parameters given,

$$V_m = 120\sqrt{2} = 169.7 \text{ V}$$

$$Z = [R^2 + (\omega L)^2]^{0.5} = 7.80\ \Omega$$

$$\theta = \tan^{-1}(\omega L/R) = 1.31 \text{ rad}$$

$$\alpha = \sin^{-1}(100/169.7) = 36.1^\circ = 0.630 \text{ rad}$$

$$\omega\tau = 377(0.02/2) = 3.77 \text{ rad}$$

(a) Using Eq. (3-22),

$$i(\omega t) = 21.8 \sin(\omega t - 1.31) - 50 + 75.3e^{-\omega t/3.77} \quad \text{A}$$

The extinction angle β is found from the solution of

$$i(\beta) = 21.8 \sin(\beta - 1.31) - 50 + 75.3e^{-\beta/3.77} = 0$$

Which results in $\beta = 3.37\ rad$ (193°) using root-finding software.

(b) Using the preceding expression for i(_t) in Eq. (3-24) and using a numerical integration program, the rms current is:

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_{0.63}^{3.37} i^2(\omega t)\,d(\omega t)} = 3.98 \text{ A}$$

Resulting in:

$$P_R = I_{\text{rms}}^2 R = 3.98^2(2) = 31.7 \text{ W}$$

(c) The power absorbed by the dc source is $I_o V_{dc}$. Using Eq. (3-26),

$$I_o = \frac{1}{2\pi}\int_{0.63}^{3.37} i(\omega t)\,d(\omega t) = 2.25 \text{ A}$$

Yielding

$$P_{\text{dc}} = I_o V_{\text{dc}} = (2.25)(100) = 225 \text{ W}$$

(d) The power supplied by the AC source is the sum of the powers absorbed by the load.

$$P_s = P_R + P_{\text{dc}} = 31.2 + 225 = 256 \text{ W}$$

The power factor is

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{s,\text{rms}} I_{\text{rms}}} = \frac{256}{(120)(3.98)} = 0.54$$

3.6 INDUCTOR-SOURCE LOAD

Using Inductance to Limit Current

Another variation of the half-wave rectifier circuit has a load that consists of an inductor and a dc source, as shown in Fig. 3-6. Although a practical implementation of this circuit would contain some resistance, the resistance may be negligible compared to other circuit parameters.

Starting at $\omega t = 0$ and assuming zero initial current in the inductor, the diode remains reverse-biased until the AC source voltage reaches the DC voltage. The value of (ωt) at which the diode starts to conduct is α, calculated using Eq. (3-18).

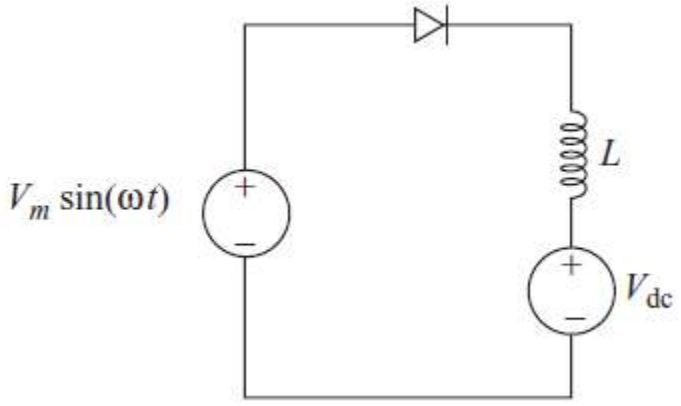


Figure 3-6 Half-wave rectifier with inductor source load.

With the diode conducting, Kirchhoff's voltage law for the circuit is:

$$V_m \sin(\omega t) = L \frac{di(t)}{dt} + V_{\text{dc}} \qquad (3\text{-}29)$$

Or

$$V_m \sin(\omega t) = \frac{L}{\omega} \frac{di(\omega t)}{dt} + V_{\text{dc}} \qquad (3\text{-}30)$$

Rearranging gives

$$\frac{di(\omega t)}{dt} = \frac{V_m \sin(\omega t) - V_{dc}}{\omega L} \qquad (3\text{-}31)$$

Solving for $i(\omega t)$,

$$i(\omega t) = \frac{1}{\omega L}\int_{\alpha}^{\omega t} V_m \sin\lambda \, d(\lambda) - \frac{1}{\omega L}\int_{\alpha}^{\omega t} V_{dc} \, d(\lambda) \qquad (3\text{-}32)$$

Performing the integration,

$$i(\omega t) = \begin{cases} \frac{V_m}{\omega L}(\cos\alpha - \cos\omega t) + \frac{V_{dc}}{\omega L}(\alpha - \omega t) & \text{for } \alpha \le \omega t \le \beta \\ 0 & \text{otherwise} \end{cases} \qquad (3\text{-}33)$$

Adistinct feature of this circuit is that the power supplied by the source is the same as that absorbed by the dc source, less any losses associated with a nonideal diode and inductor. If the objective is to transfer power from the ac source to the DC source, losses are kept to a minimum by using this circuit.

Example 3-6(Half-Wave Rectifier with Inductor-Source Load)
For the circuit of Fig. 3-6, the AC source is 120 V rms at 60 Hz, $L = 50\ mH$, and $V_{dc} = 72\ V$.
Determine (a) an expression for the current, (b) the power absorbed by the DC source, and (c) the power factor.
■ Solution
For the parameters given,

$$\alpha = \sin^{-1}\left(\frac{72}{120\sqrt{2}}\right) = 25.1° = 0.438 \text{ rad}$$

(a) The equation for current is found from Eq. (3-33).

$$i(\omega t) = 9.83 - 9.00\cos(\omega t) - 3.82\,\omega t \quad \text{A} \quad \text{for } \alpha \le \omega t \le \beta$$

Where $(\beta_)$ is found to be 4.04 rad from the numerical solution of $9.83 - \cos\beta - 3.82\beta = 0$.

(b) The power absorbed by the dc source is $I_o V_{dc}$, where:

$$I_o = \frac{1}{2\pi}\int_{\alpha}^{\beta} i(\omega t)\, d(\omega t)$$

$$= \frac{1}{2\pi}\int_{0.438}^{4.04} [9.83 - 9.00\cos(\omega t) - 3.82\ \omega t]\, d(\omega t) = 2.46\ \text{A}$$

resulting in

$$P_{\text{dc}} = V_{\text{dc}}\, I_o = (2.46)(72) = 177\ \text{W}$$

(c) The rms current is found from

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\beta} i^2(\omega t)\, d(\omega t)} = 3.81\ \text{A}$$

Therefore,

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{\text{rms}}\, I_{\text{rms}}} = \frac{177}{(120)(3.81)} = 0.388$$

3.7 THE FREEWHEELING DIODE

Creating a DC Current

A freewheeling diode, D_2 in Fig. 3-7*a*, can be connected across an *RL* load as shown. The behavior of this circuit is somewhat different from that of the half-wave rectifier of Fig. 3-2. The key to the analysis of this circuit is to determine when each diode conducts. First, it is observed that both diodes cannot be forward-biased at the same time. Kirchhoff's voltage law around the path containing the source and the two diodes shows that one diode must be reverse-biased.

Diode D_1 will be on when the source is positive, and diode D_2 will be on when the source is negative.

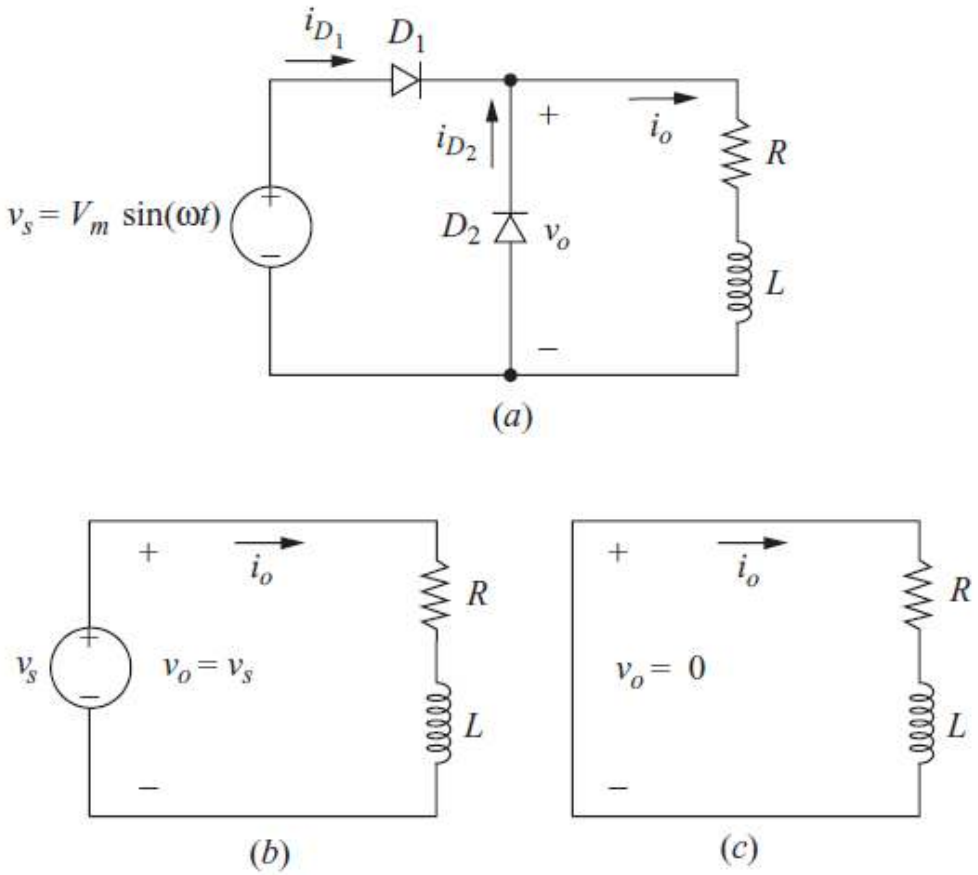


Figure 3-7 (*a*) Half-wave rectifier with freewheeling diode; (*b*) Equivalent circuit for $v_S > 0$; (*c*) Equivalent circuit for $v_S < 0$.

For a positive source voltage,

• D_1 is on.

• D_2 is off.

• The equivalent circuit is the same as that of Fig. 3-2, shown again in Fig. 3-7*b*.

• The voltage across the *RL* load is the same as the source.

For a negative source voltage,

• D_1 is off.

• D_2 is on.

• The equivalent circuit is the same at that of Fig. 3-7*c*.

• The voltage across the *RL* load is zero.

Since the voltage across the *RL* load is the same as the source voltage when the source is positive and is zero when the source is negative, the load voltage is a half-wave rectified sine wave.

When the circuit is first energized, the load current is zero and cannot change instantaneously. The current reaches periodic steady state after a few periods (depending on the *L*/*R* time constant), which means that the current

at the end of a period is the same as the current at the beginning of the period, as shown in Fig. 3-8. The steady-state current is usually of greater interest than the transient that occurs when the circuit is first energized. Steady-state load, source, and diode currents are shown in Fig. 3-9.

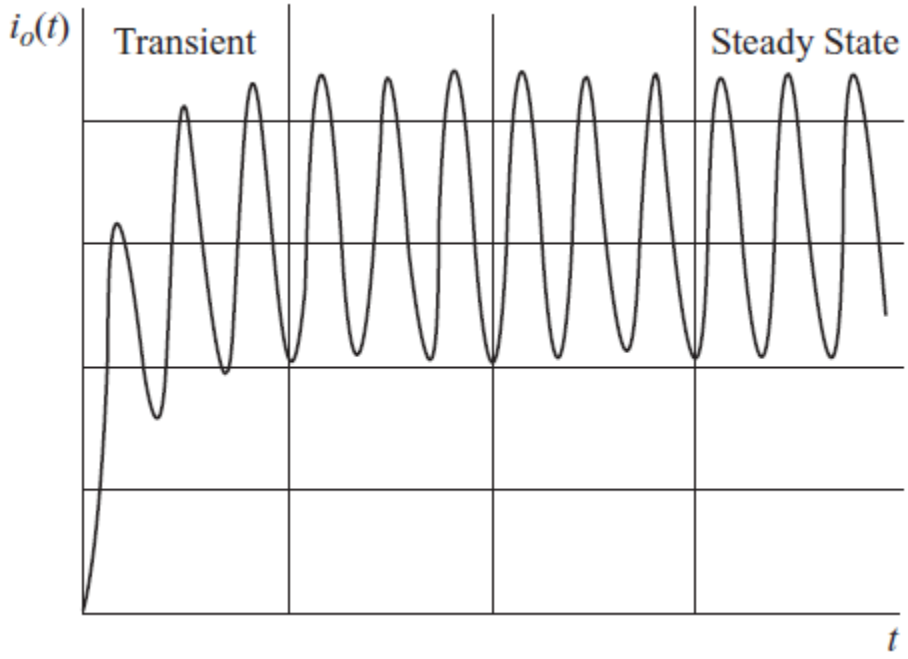


Figure 3-8 Load current reaching steady state after the circuit is energized.

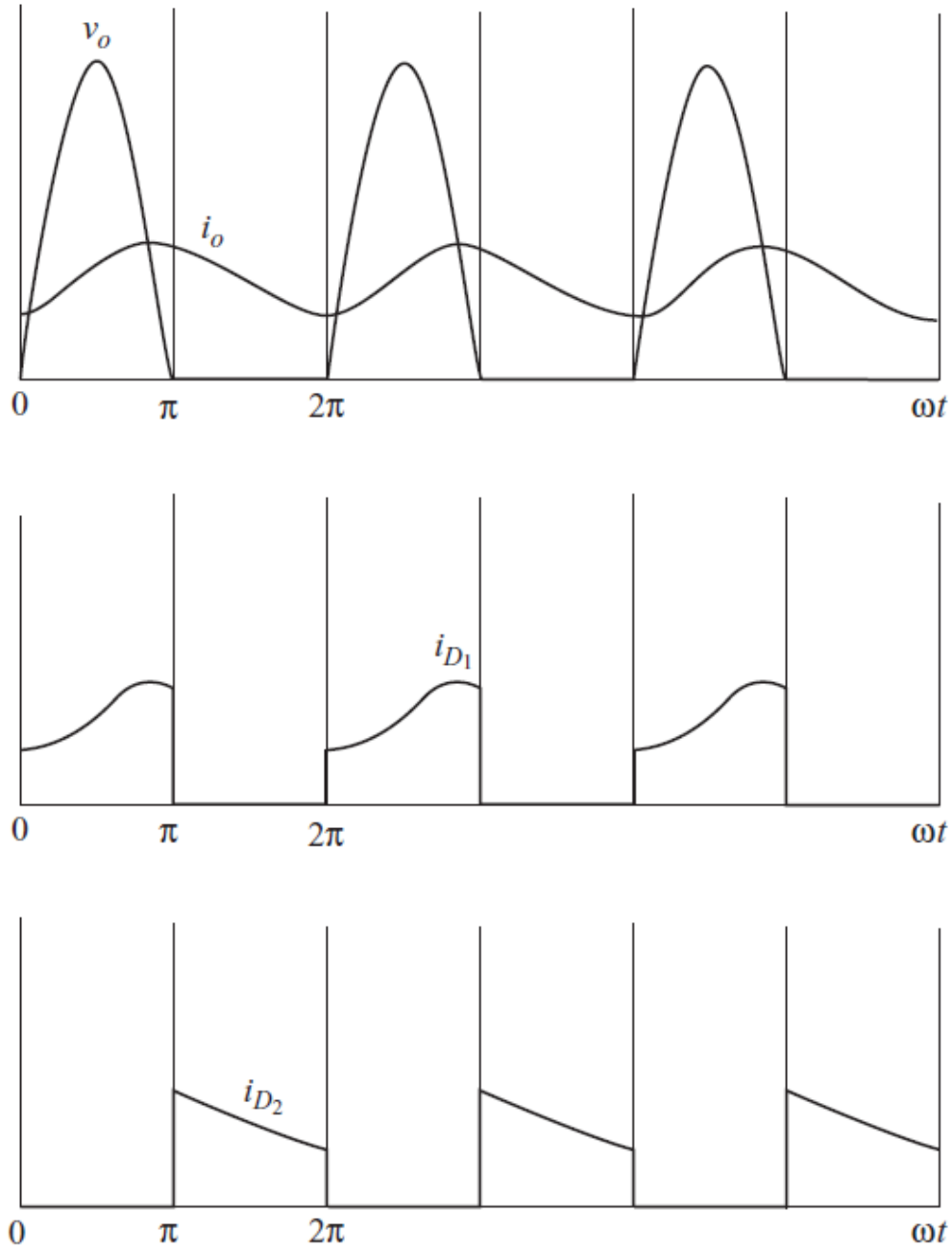


Figure 3-9 Steady-state load voltage and current waveforms with freewheeling diode.

The Fourier series for the half-wave rectified sine wave for the voltage across the load is:

$$v(t) = \frac{V_m}{\pi} + \frac{V_m}{2}\sin(\omega_0 t) - \sum_{n=2,4,6\ldots}^{\infty} \frac{2V_m}{(n^2-1)\pi}\cos(n\omega_0 t) \qquad (3\text{-}34)$$

The current in the load can be expressed as a Fourier series by using superposition, taking each frequency separately. The Fourier series method is illustrated in Example 3-7.

Example 3-7(Half-Wave Rectifier with Freewheeling Diode)
Determine the average load voltage and current, and determine the power absorbed by the resistor in the circuit of Fig. 3-7a, where $R = 2\ \Omega$ and $L = 25\ mH, V_m$ is 100 V, and the frequency is 60 Hz.

■ **Solution**
The Fourier series for this half-wave rectified voltage that appears across the load is obtained from Eq. (3-34). The average load voltage is the DC term in the Fourier series:

$$V_o = \frac{V_m}{\pi} = \frac{100}{\pi} = 31.8\ \text{V}$$

Average load current is

$$I_o = \frac{V_o}{R} = \frac{31.8}{2} = 15.9\ \text{A}$$

Load power can be determined from $I_{rms}^2\ R$, and rms current is determined from the Fourier components of current. The amplitudes of the AC current components are determined from phasor analysis:

$$I_n = \frac{V_n}{Z_n}$$

where $$Z_n = |R + jn\omega_0 L| = |2 + jn377(0.025)|$$

The AC voltage amplitudes are determined from Eq. (3-34), resulting in:

$$V_1 = \frac{V_m}{2} = \frac{100}{2} = 50\ \text{V}$$

$$V_2 = \frac{2V_m}{(2^2 - 1)\pi} = 21.2\ \text{V}$$

$$V_4 = \frac{2V_m}{(4^2 - 1)\pi} = 4.24\ \text{V}$$

$$V_6 = \frac{2V_m}{(6^2 - 1)\pi} = 1.82\ \text{V}$$

The resulting Fourier terms are as follows:

n	V_n (V)	Z_n (Ω)	I_n (A)
0	31.8	2.00	15.9
1	50.0	9.63	5.19
2	21.2	18.96	1.12
4	4.24	37.75	0.11
6	1.82	56.58	0.03

The rms current is obtained using Eq. (2-64).

$$I_{\text{rms}} = \sqrt{\sum_{k=0}^{\infty} I_{k,\text{rms}}} \approx \sqrt{15.9^2 + \left(\frac{5.19}{\sqrt{2}}\right)^2 + \left(\frac{1.12}{\sqrt{2}}\right)^2 + \left(\frac{0.11}{\sqrt{2}}\right)^2} = 16.34 \text{ A}$$

Notice that the contribution to rms current from the harmonics decreases as n increases, and higher-order terms are not significant. Power in the resistor is $I_{rms}^2 R = (16.34)^2 \times 2 = 534\, W$.

Reducing Load Current Harmonics

The average current in the *RL* load is a function of the applied voltage and the resistance but not the inductance. The inductance affects only the ac terms in the Fourier series. If the inductance is infinitely large, the impedance of the load to AC terms in the Fourier series is infinite, and the load current is purely DC. The load current is then:

$$i_o(t) \approx I_o = \frac{V_o}{R} = \frac{V_m}{\pi R} \qquad \frac{L}{R} \to \infty \tag{3-35}$$

Alarge inductor ($L/R \gg T$) with a freewheeling diode provides a means of establishing a nearly constant load current. Zero-to-peak fluctuation in load current can be estimated as being equal to the amplitude of the first ac term in the Fourier series. The peak-to-peak ripple is then:

$$\Delta I_o \approx 2I_1 \tag{3-36}$$

Example 3-8 (Half-Wave Rectifier with Freewheeling Diode: $L/R \to \infty$)
For the half-wave rectifier with a freewheeling diode and *RL* load as shown in Fig. 3-7*a*, the source is $240\ V\ rms$ at 60 Hz and $R = 8\ \Omega$. (*a*) Assume *L* is infinitely large. Determine the power absorbed by the load and the power factor as seen by the source. Sketch v_o, i_{D1} , and i_{D2}.
(*b*) Determine the average current in each diode.

(*c*) For a finite inductance, determine *L* such that the peak-to-peak current is no more than 10 percent of the average current.

■ **Solution**

(*a*) The voltage across the *RL* load is a half-wave rectified sine wave, which has an average value of V_m/π. The load current is:

$$i(\omega t) = I_o = \frac{V_o}{R} = \frac{V_m/\pi}{R} = \frac{\left(240\sqrt{2}\right)/\pi}{8} = 13.5 \text{ A} \approx I_{\text{rms}}$$

Power in the resistor is:

$$P = (I_{\text{rms}})^2 R = (13.5)^2 8 = 1459 \text{ W}$$

The source rms current is computed from:

$$I_{s,\text{rms}} = \sqrt{\frac{1}{2\pi}\int_0^{\pi} (13.5)^2 d(\omega t)} = 9.55 \text{ A}$$

The power factor is:

$$\text{pf} = \frac{P}{V_{s,\text{rms}} I_{s,\text{rms}}} = \frac{1459}{(240)(9.55)} = 0.637$$

Voltage and current waveforms are shown in Fig. 3-10.

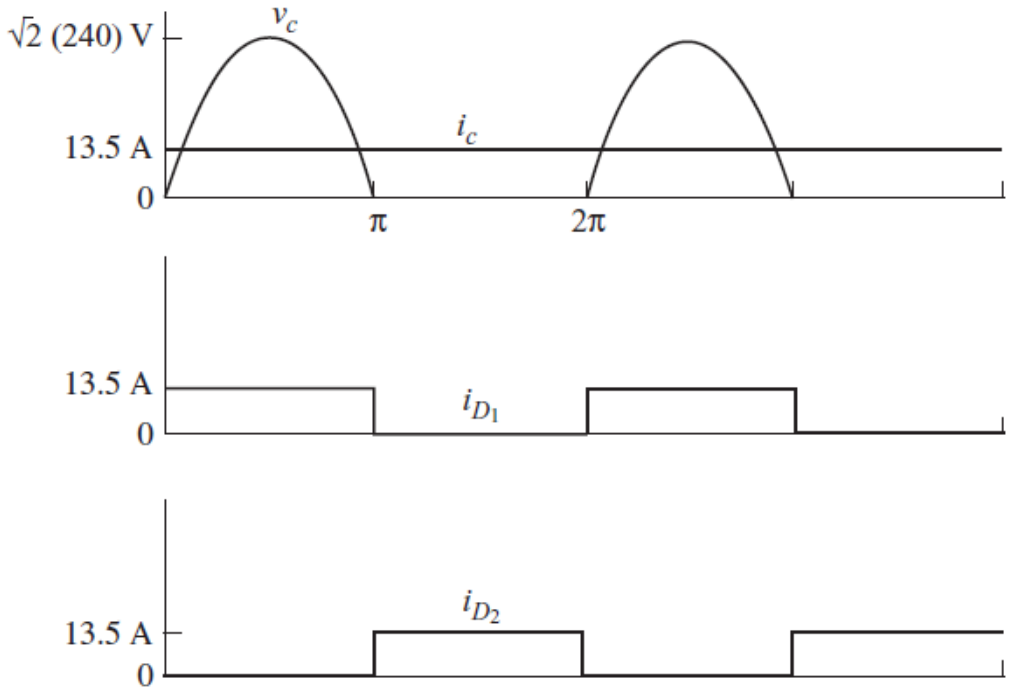


Figure 3-10 Waveforms for the half-wave rectifier with freewheeling diode of Example 3-8 with $L/R \rightarrow \infty$.

(b) Each diode conducts for one-half of the time. Average current for each diode is $I_o/_2 = \frac{13.5}{2} = 6.75\ A$.

(*c*) The value of inductance required to limit the variation in load current to 10 percent can be approximated from the fundamental frequency of the Fourier series. The voltage input to the load for $n = 1$ in Eq. (3-34) has amplitude:

$$\frac{V_m}{2} = \frac{\sqrt{2}(240)}{2} = 170\ V$$

The peak-to-peak current must be limited to:

$$\Delta I_o = (0.10)(I_o) = (0.10)(13.5) = 1.35\ \text{A}$$

Which corresponds to an amplitude of $1.35/_2 = 0.675\ A$. The load impedance at the fundamental frequency must then be:

$$Z_1 = \frac{V_1}{I_1} = \frac{170}{0.675} = 251\ \Omega$$

The load impedance is

$$Z_1 = 251 = |R + j\omega L| = |8 + j377L|$$

Since the 8 Ω resistance is negligible compared to the total impedance, the inductance can be approximated as:

$$L \approx \frac{Z_1}{\omega} = \frac{251}{377} = 0.67\ \text{H}$$

The inductance will have to be slightly larger than 0.67 H because Fourier terms higher than $n = 1$ were neglected in this estimate.

3.8 HALF-WAVE RECTIFIER WITH A CAPACITOR FILTER

Creating a DC Voltage from an AC Source

A common application of rectifier circuits is to convert an ac voltage input to a DC voltage output. The half-wave rectifier of Fig. 3-11*a* has a parallel *RC* load.

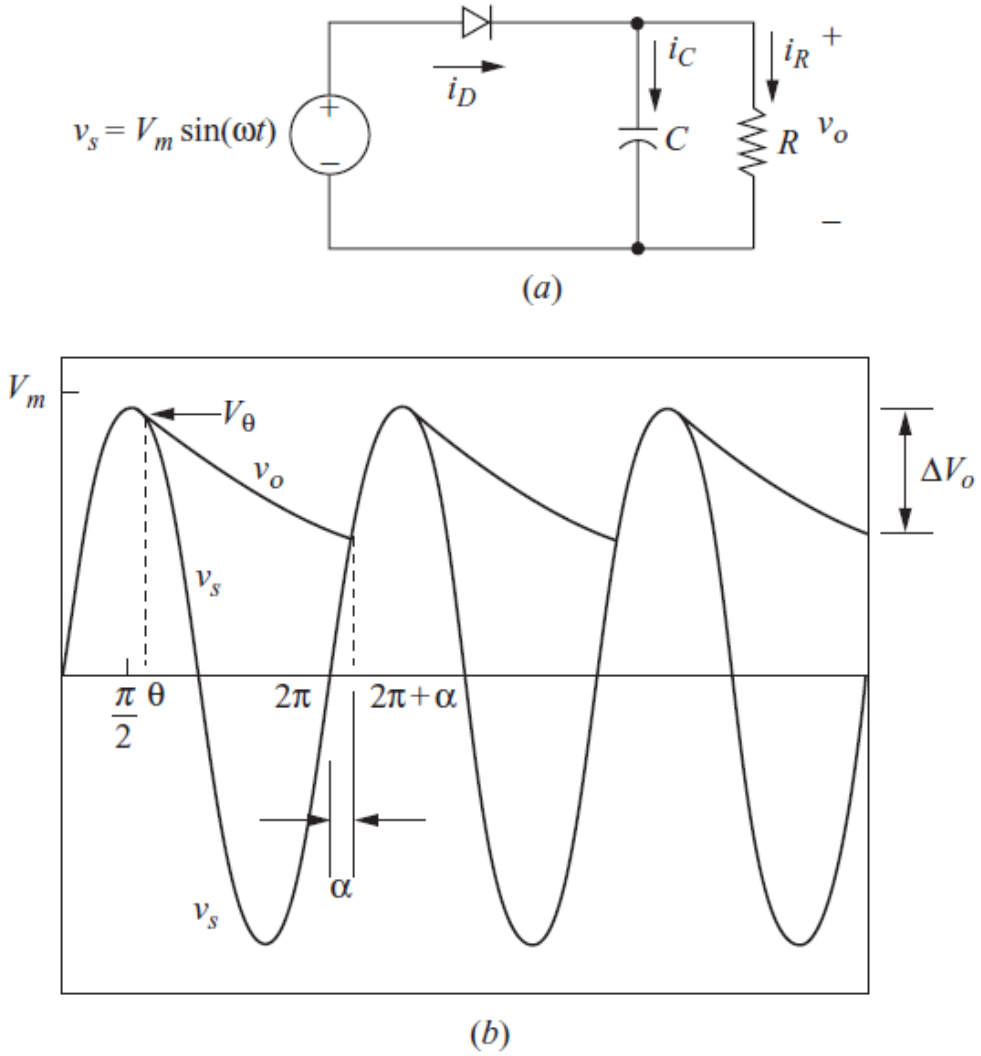


Figure 3-11 (*a*) Half-wave rectifier with *RC* load; (*b*) Input and output voltages.

The purpose of the capacitor is to reduce the variation in the output voltage, making it more like DC. The resistance may represent an external load, and the capacitor may be a filter which is part of the rectifier circuit.

Assuming the capacitor is initially uncharged and the circuit is energized at $\omega t = 0$, the diode becomes forward-biased as the source becomes positive. With the diode on, the output voltage is the same as the source voltage, and the capacitor charges. The capacitor is charged to V_m when the input voltage reaches its positive peak at $\omega t = \pi/2$.

As the source decreases after $\omega t = \pi/2$, the capacitor discharges into the load resistor. At some point, the voltage of the source becomes less than the output voltage, reverse-biasing the diode and isolating the load from the

source. The output voltage is a decaying exponential with time constant *RC* while the diode is off.

The point when the diode turns off is determined by comparing the rates of change of the source and the capacitor voltages. The diode turns off when the downward rate of change of the source exceeds that permitted by the time constant of the RC load. The angle $\omega t = \theta$ is the point when the diode turns off in Fig. 3-11b. The output voltage is described by:

$$v_o(\omega t) = \begin{cases} V_m \sin \omega t & \text{diode on} \\ V_\theta e^{-(\omega t - \theta)/\omega RC} & \text{diode off} \end{cases} \tag{3-37}$$

Where

$$V_\theta = V_m \sin \theta \tag{3-38}$$

The slopes of these functions are

$$\frac{d}{d(\omega t)}[V_m \sin(\omega t)] = V_m \cos(\omega t) \tag{3-39}$$

And

$$\frac{d}{d(\omega t)}\left(V_m \sin\theta\, e^{-(\omega t - \theta)/\omega RC}\right) = V_m \sin\theta\left(-\frac{1}{\omega RC}\right)e^{-(\omega t - \theta)/\omega RC} \tag{3-40}$$

At $\omega t = \theta$, the slopes of the voltage functions are equal:

$$V_m \cos\theta = \left(\frac{V_m \sin\theta}{-\omega RC}\right)e^{-(\theta-\theta)/\omega RC} = \frac{V_m \sin\theta}{-\omega RC}$$

$$\frac{V_m \cos\theta}{V_m \sin\theta} = \frac{1}{-\omega RC}$$

$$\frac{1}{\tan\theta} = \frac{1}{-\omega RC}$$

Solving for θ and expressing θ so it is in the proper quadrant, we have:

$$\boxed{\theta = \tan^{-1}(-\omega RC) = -\tan^{-1}(\omega RC) + \pi} \tag{3-41}$$

In practical circuits where the time constant is large,

$$\theta \approx \frac{\pi}{2} \quad \text{and} \quad V_m \sin\theta \approx V_m \tag{3-42}$$

When the source voltage comes back up to the value of the output voltage in the next period, the diode becomes forward-biased, and the output again is the same as the source voltage. The angle at which the diode turns on in the second period, $\omega t = 2\pi + \alpha$, is the point when the sinusoidal source reaches the same value as the decaying exponential output:

$$V_m \sin(2\pi + \alpha) = (V_m \sin\theta) e^{-(2\pi+\alpha-\theta)/\omega RC}$$

Or

$$\sin\alpha - (\sin\theta) e^{-(2\pi+\alpha-\theta)/\omega RC} = 0 \tag{3-43}$$

Equation (3-43) must be solved numerically for α.

The current in the resistor is calculated from $i_R = \frac{v_o}{R}$.

The current in the capacitor is calculated from:

$$i_C(t) = C\frac{dv_o(t)}{dt}$$

Which can also be expressed, using _t as the variable, as

$$i_C(\omega t) = \omega C\frac{dv_o(\omega t)}{d(\omega t)}$$

Using v_o from Eq. (3-37),

$$i_C(\omega t) = \begin{cases} -\left(\frac{V_m \sin\theta}{R}\right) e^{-(\omega t-\theta)/\omega RC} & \text{for } \theta \le \omega t \le 2\pi + \alpha \quad \text{(diode off)} \\ \omega C V_m \cos(\omega t) & \text{for } 2\pi + \alpha \le \omega t \le 2\pi + \theta \quad \text{(diode on)} \end{cases} \tag{3-44}$$

The source current, which is the same as the diode current, is:

$$i_S = i_D = i_R + i_C \tag{3-45}$$

The average capacitor current is zero, so the average diode current is the same as the average load current. Since the diode is on for a short time in each cycle, the peak diode current is generally much larger than the average diode current.

Peak capacitor current occurs when the diode turns on at $\omega t = 2\pi + \alpha$. From Eq. (3-44),

$$I_{C,\text{peak}} = \omega C V_m \cos(2\pi + \alpha) = \omega C V_m \cos\alpha \tag{3-46}$$

Resistor current at $\omega t = 2\pi + \alpha$ is obtained from Eq. (3-37).

$$i_R(2\omega t + \alpha) = \frac{V_m \sin(2\omega t + \alpha)}{R} = \frac{V_m \sin\alpha}{R} \tag{3-47}$$

Peak diode current is:

$$I_{D,\text{peak}} = \omega C V_m \cos\alpha + \frac{V_m \sin\alpha}{R} = V_m\left(\omega C \cos\alpha + \frac{\sin\alpha}{R}\right) \tag{3-48}$$

The effectiveness of the capacitor filter is determined by the variation in output voltage. This may be expressed as the difference between the maximum and minimum output voltage, which is the peak-to-peak ripple voltage. For the halfwave rectifier of Fig. 3-11a, the maximum output voltage is (V_m). The minimum output voltage occurs at $\omega t = 2\pi + \alpha$, which can be computed from $V_m \sin\alpha$. The peak-to-peak ripple for the circuit of Fig. 3-11a is expressed as:

$$\Delta V_o = V_m - V_m \sin\alpha = V_m(1 - \sin\alpha) \tag{3-49}$$

In circuits where the capacitor is selected to provide for a nearly constant DC output voltage, the *RC* time constant is large compared to the period of the sine wave, and Eq. (3-42) applies. Moreover, the diode turns on close to the peak of the sine wave when $\alpha \approx \pi/2$. The change in output voltage when the diode is off is described in Eq. (3-37). In Eq. (3-37), if $V_\theta \approx V_m$ and $\theta \approx \pi/2$, then Eq. (3-37) evaluated at $\alpha = \pi/2$ is:

$$v_o(2\pi + \alpha) = V_m e^{-(2\pi + \pi/2 - \pi/2)\omega RC} = V_m e^{-2\pi/\omega RC}$$

The ripple voltage can then be approximated as

$$\Delta V_o \approx V_m - V_m e^{-2\pi/\omega RC} = V_m\left(1 - e^{-2\pi/\omega RC}\right) \tag{3-50}$$

Furthermore, the exponential in the above equation can be approximated by the series expansion:

$$e^{-2\pi/\omega RC} \approx 1 - \frac{2\pi}{\omega RC}$$

Substituting for the exponential in Eq. (3-50), the peak-to-peak ripple is approximately

$$\Delta V_o \approx V_m\left(\frac{2\pi}{\omega RC}\right) = \frac{V_m}{fRC} \tag{3-51}$$

The output voltage ripple is reduced by increasing the filter capacitor C. As C increases, the conduction interval for the diode decreases. Therefore, increasing the capacitance to reduce the output voltage ripple results in a larger peak diode current.

Example 3-9 (Half-Wave Rectifier with RC Load)

The half-wave rectifier of Fig. 3-11a has a $120V\ rms$ source at 60 Hz, $R = 500\ \Omega, and\ C = 100\ F$. Determine (a) an expression for output voltage, (b) the peak-to-peak voltage variation on the output, (c) an expression for capacitor current, (d) the peak diode current, and (e) the value of C such that ΔV_o is 1 percent of V_m.

■ Solution

From the parameters given,

$$V_m = 120\sqrt{2} = 169.7\ \text{V}$$

$$\omega RC = (2\pi 60)(500)(10)^{-6} = 18.85\ \text{rad}$$

The angle θ is determined from Eq. (3-41).

$$\theta = -\tan^{-1}(18.85) + \pi = 1.62\,\text{rad} = 93°$$

$$V_m \sin\theta = 169.5\ \text{V}$$

The angle α is determined from the numerical solution of Eq. (3-43).

$$\sin\alpha - \sin(1.62)e^{-(2\pi+\alpha-1.62/18.85)} = 0$$

Yielding:

$$\alpha = 0.843\ \text{rad} = 48°$$

(a) Output voltage is expressed from Eq. (3-37).

$$v_o(\omega t) = \begin{cases} 169.7 \sin(\omega t) & 2\pi + \alpha \le \omega t \le 2\pi + \theta \\ 169.5e^{-(\omega t - 1.62)/18.85} & \theta \le \omega t \le 2\pi + \alpha \end{cases}$$

(b) Peak-to-peak output voltage is described by Eq. (3-49).

$$\Delta V_o = V_m(1 - \sin\alpha) = 169.7(1 - \sin 0.843) = 43 \text{ V}$$

(c) The capacitor current is determined from Eq. (3-44).

$$i_C(\omega t) = \begin{cases} -0.339e^{-(\omega t - 1.62)/18.85} \text{ A} & \theta \le \omega t \le 2\pi + \alpha \\ 6.4\cos(\omega t) \text{ A} & 2\pi + \alpha \le \omega t \le 2\pi + \theta \end{cases}$$

(d) Peak diode current is determined from Eq. (3-48).

$$I_{D,\text{peak}} = \sqrt{2}(120)\left[377(10)^{-4}\cos 0.843 + \frac{\sin 8.43}{500}\right]$$

$$= 4.26 + 0.34 = 4.50 \text{ A}$$

(e) For $\Delta V_o = 0.01\, V_m$, Eq. (3-51) can be used.

$$C \approx \frac{V_m}{fR(\Delta V_o)} = \frac{V_m}{(60)(500)(0.01 V_m)} = \frac{1}{300}\text{F} = 3333\ \mu\text{F}$$

Note that peak diode current can be determined from Eq. (3-48) using an estimate of α from Eq. (3-49).

$$\alpha \approx \sin^{-1}\left(1 - \frac{\Delta V_o}{V_m}\right) = \sin^{-1}\left(1 - \frac{1}{fRC}\right) = 81.9°$$

From Eq. (3-48), peak diode current is 30.4 A.

3.9 THE CONTROLLED HALF-WAVE RECTIFIER

The half-wave rectifiers analyzed previously in this chapter are classified as uncontrolled rectifiers. Once the source and load parameters are established, the DC level of the output and the power transferred to the load are fixed quantities.

A way to control the output of a half-wave rectifier is to use an SCR instead of a diode. Figure 3-13*a* shows a basic controlled half-wave rectifier with a resistive load. Two conditions must be met before the SCR can conduct:

1. The SCR must be forward-biased ($v_{SCR} > 0$).
2. A current must be applied to the gate of the SCR.

Unlike the diode, the SCR will not begin to conduct as soon as the source becomes positive. Conduction is delayed until a gate current is applied, which is the basis for using the SCR as a means of control. Once the SCR is conducting, the gate current can be removed and the SCR remains on until the current goes to zero.

Resistive Load

Figure 3-13*b* shows the voltage waveforms for a controlled half-wave rectifier with a resistive load. Agate signal is applied to the SCR at $\omega t = \alpha$, where (α) is the delay angle.

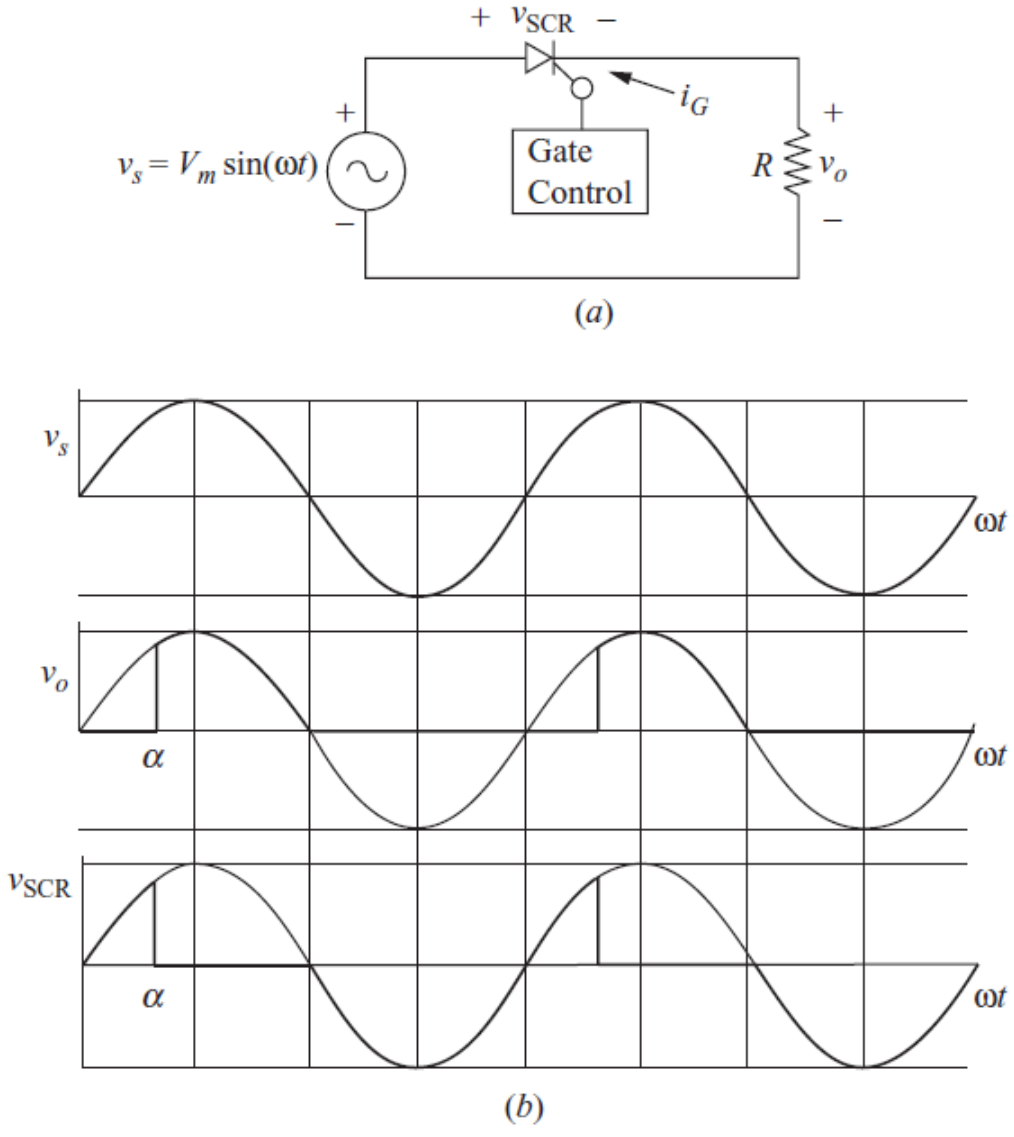


Figure 3-13 (*a*) A basic controlled rectifier; (*b*) Voltage waveforms.

The average (DC) voltage across the load resistor in Fig. 3-13*a* is:

$$V_o = \frac{1}{2\pi}\int_{\alpha}^{\pi} V_m \sin(\omega t)\, d(\omega t) = \frac{V_m}{2\pi}(1 + \cos\alpha) \tag{3-52}$$

The power absorbed by the resistor is V_{rms}^2/R, where the rms voltage across the resistor is computed from:

$$V_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_{0}^{2\pi} v_o^2(\omega t)\, d(\omega t)}$$

$$= \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\pi} [V_m \sin(\omega t)]^2\, d(\omega t)}$$

$$= \frac{V_m}{2}\sqrt{1 - \frac{\alpha}{\pi} + \frac{\sin(2\alpha)}{2\pi}} \tag{3-53}$$

Example 3-10 (Controlled Half-Wave Rectifier with Resistive Load)
Design a circuit to produce an average voltage of 40 V across a $100\ \Omega$ load resistor from a $120\ V\ rms\ 60\ Hz$ AC source. Determine the power absorbed by the resistance and the power factor.

■ **Solution**

If an uncontrolled half-wave rectifier is used, the average voltage will be $\frac{V_m}{\pi} = \frac{\sqrt{2}\times 120}{\pi} = 54\ V$.

Some means of reducing the average resistor voltage to the design specification of 40 V must be found. A series resistance or inductance could be added to an uncontrolled rectifier, or a controlled rectifier could be used. The controlled rectifier of Fig. 3-13*a* has the advantage of not altering the load or introducing losses, so it is selected for this application.

Equation (3-52) is rearranged to determine the required delay angle:

$$\alpha = \cos^{-1}\left[V_o\left(\frac{2\pi}{V_m}\right) - 1\right]$$

$$= \cos^{-1}\left\{40\left[\frac{2\pi}{\sqrt{2}(120)}\right] - 1\right\} = 61.2° = 1.07 \text{ rad}$$

Equation (3-53) gives

$$V_{\text{rms}} = \frac{\sqrt{2}(120)}{2}\sqrt{1 - \frac{1.07}{\pi} + \frac{\sin[2(1.07)]}{2\pi}} = 75.6 \text{ V}$$

Load power is

$$P_R = \frac{V_{\text{rms}}^2}{R} = \frac{(75.6)^2}{100} = 57.1 \text{ W}$$

The power factor of the circuit is

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{S,\text{rms}} I_{\text{rms}}} = \frac{57.1}{(120)(75.6/100)} = 0.63$$

***RL* Load**

A controlled half-wave rectifier with an *RL* load is shown in Fig. 3-14*a*. The analysis of this circuit is similar to that of the uncontrolled rectifier.

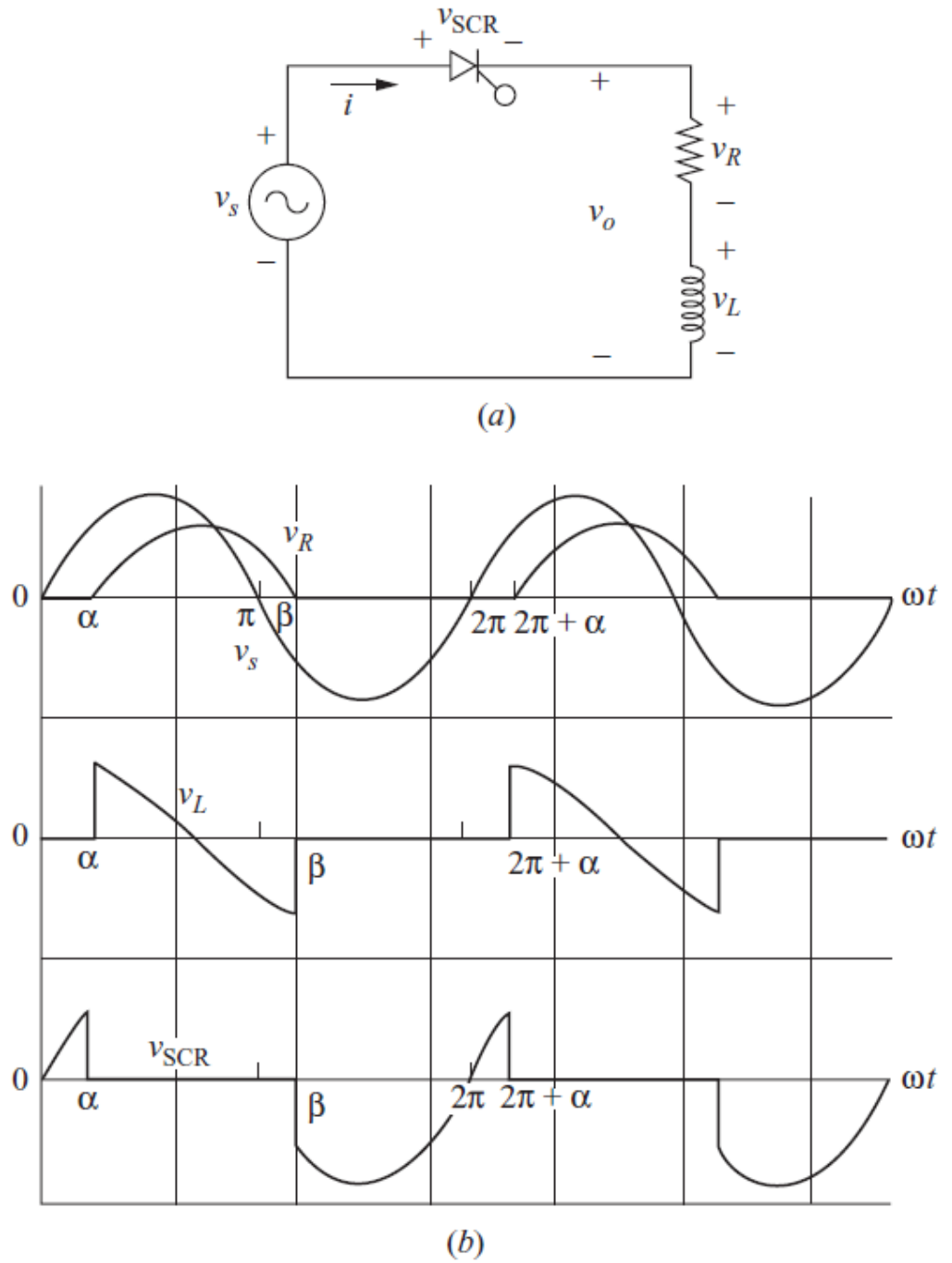


Figure 3-14 (*a*) Controlled half-wave rectifier with *RL* load; (*b*) Voltage waveforms.

The current is the sum of the forced and natural responses, and Eq. (3-9) applies:

$$i(\omega t) = i_f(\omega t) + i_n(\omega t) = \frac{V_m}{Z} \sin(\omega t - \theta) + Ae^{-\omega t/\omega\tau}$$

The constant A is determined from the initial condition $i(\alpha) = 0$:

$$i(\alpha) = 0 = \frac{V_m}{Z}\sin(\alpha - \theta) + Ae^{-\alpha/\omega\tau}$$

$$A = \left[-\frac{V_m}{Z}\sin(\alpha - \theta)\right] = e^{\alpha/\omega\tau} \tag{3-54}$$

Substituting for A and simplifying,

$$i(\omega t) = \begin{cases} \frac{V_m}{Z}\left[\sin(\omega t - \theta) - \sin(\alpha - \theta)e^{(\alpha-\omega t)/\omega\tau}\right] & \text{for } \alpha \le \omega t \le \beta \\ 0 & \text{otherwise} \end{cases} \tag{3-55}$$

The extinction angle β is defined as the angle at which the current returns to zero, as in the case of the uncontrolled rectifier. When $\omega t = \beta$,

$$i(\beta) = 0 = \frac{V_m}{Z}\left[\sin(\beta - \theta) - \sin(\alpha - \theta)e^{(\alpha-\beta)/\omega\tau}\right] \tag{3-56}$$

Which must be solved numerically for β. The angle $(\beta - \alpha)$is called the *conduction angle* γ.

Figure 3-14*b* shows the voltage waveforms.

The average (DC) output voltage is:

$$V_o = \frac{1}{2\pi}\int_{\alpha}^{\beta} V_m \sin(\omega t)\,d(\omega t) = \frac{V_m}{2\pi}(\cos\alpha - \cos\beta) \tag{3-57}$$

The average current is computed from:

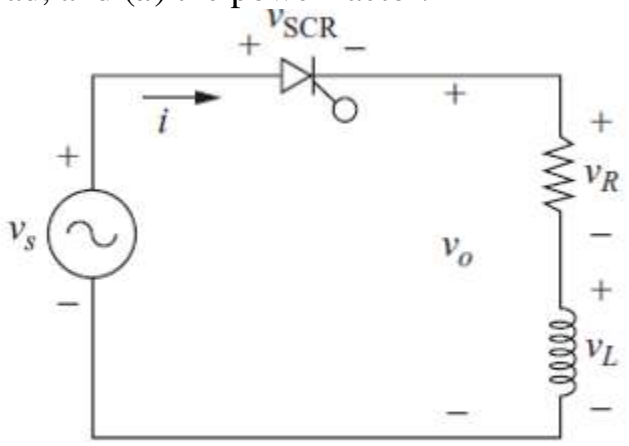
$$I_o = \frac{1}{2\pi}\int_{\alpha}^{\beta} i(\omega t)\,d(\omega t) \tag{3-58}$$

Where $i(\omega t)$ is defined in Eq. (3-55).

Power absorbed by the load is $I_{rms}^2 R$, where the rms current is computed from:

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\beta} i^2(\omega t)\,d(\omega t)} \qquad (3\text{-}59)$$

Example 3-11 (Controlled Half-Wave Rectifier with *RL* Load)
For the circuit of Fig. 3-14*a*, the source is $120\ V\ rms$ at $60\ Hz$, $R = 20\ \Omega, L = 0.04\ H$, and the delay angle is 45°.
Determine (*a*) an expression for $i(\omega t)$, (*b*) the average current, (*c*) the power absorbed by the load, and (*d*) the power factor.



(*a*)

■ **Solution**
(*a*) From the parameters given,

$$V_m = 120\sqrt{2} = 169.7\ \text{V}$$
$$Z = [R^2 + (\omega L)^2]^{0.5} = [20^2 + (377{*}0.04)^2]^{0.5} = 25.0\ \Omega$$
$$\theta = \tan^{-1}(\omega L/R) = \tan^{-1}(377{*}0.04/20) = 0.646\ \text{rad}$$
$$\omega\tau = \omega L/R = 377{*}0.04/20 = 0.754$$
$$\alpha = 45° = 0.785\ \text{rad}$$

Substituting the preceding quantities into Eq. (3-55), current is expressed as

$$i(\omega t) = 6.78 \sin(\omega t - 0.646) - 2.67e^{-\omega t/0.754} \quad \text{A} \quad \text{for } \alpha \le \omega t \le \beta$$

The preceding equation is valid from α to β, where β is found numerically by setting the equation to zero and solving for ωt, with the result $\beta = 3.79\ rad$ (217°).
The conduction angle is $\gamma = \beta - \alpha = 3.79 - 0.785 = 3.01\ rad = 172°$.

(b) Average current is determined from Eq. (3-58).

$$I_o = \frac{1}{2\pi}\int_{0.785}^{3.79}\left[6.78\sin(\omega t - 0.646) - 2.67e^{-\omega t/0.754}\right]d(\omega t) = 2.19 \text{ A}$$

(c)The power absorbed by the load is computed from $I_{rms}^2 R$, where:

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_{0.785}^{3.79}\left[6.78\sin(\omega t - 0.646) - 2.67e^{-\omega t/0.754}\right]^2 d(\omega t)} = 3.26 \text{ A}$$

yielding

$$P = I_{\text{rms}}^2 R = (3.26)^2(20) = 213 \text{ W}$$

(d) The power factor is:

$$\text{pf} = \frac{P}{S} = \frac{213}{(120)(3.26)} = 0.54$$

RL-Source Load

A controlled rectifier with a series resistance, inductance, and a DC source is shown in Fig. 3-15.

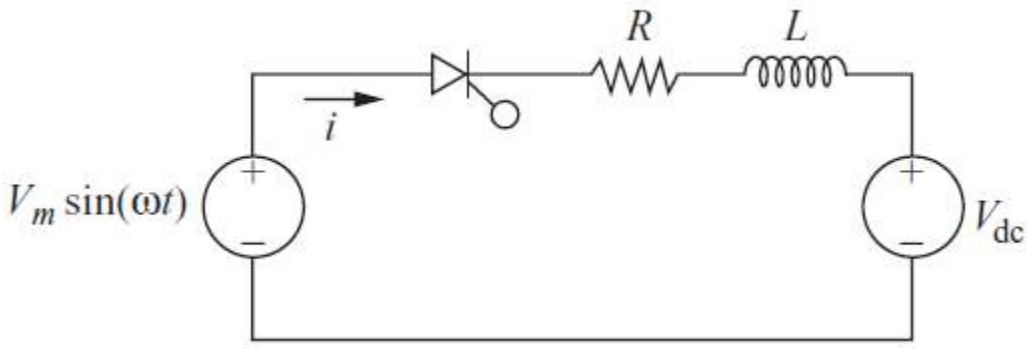


Figure 3-15 Controlled rectifier with *RL*-source load.

The analysis of this circuit is very similar to that of the uncontrolled half-wave rectifier discussed earlier in this chapter.

The major difference is that for the uncontrolled rectifier, conduction begins as soon as the source voltage reaches the level of the dc voltage. For the controlled rectifier, conduction begins when a gate signal is applied to the SCR, provided the SCR is forward-biased.

Thus, the gate signal may be applied at any time that the ac source is larger than the DC source:

$$\alpha_{\min} = \sin^{-1}\left(\frac{V_{\text{dc}}}{V_m}\right) \tag{3-60}$$

Current is expressed as in Eq. (3-22), with α specified within the allowable range:

$$i(\omega t)\begin{cases} \dfrac{V_m}{Z}\sin(\omega t - \theta) - \dfrac{V_{\text{dc}}}{R} + Ae^{-\omega t/\omega\tau} & \text{for } \alpha \le \omega t \le \beta \\ 0 & \text{otherwise} \end{cases} \tag{3-61}$$

where A is determined from Eq. (3-61):

$$A = \left[-\frac{V_m}{Z}\sin(\alpha - \theta) + \frac{V_{\text{dc}}}{R}\right]e^{\alpha/\omega\tau}$$

Example 3-12 (Controlled Rectifier with *RL*-Source Load)
The controlled half-wave rectifier of Fig. 3-15 has an AC input of $120\ V\ rms$ at $60\ Hz$, $R = 2\Omega$, L $= 20\ mH$, and $V_{dc} = 100\ V$. The delay angle α is 45°. Determine (*a*) an expression for the current, (*b*) the power absorbed by the resistor, and (*c*) the power absorbed by the DC source in the load.

■ **Solution:**
From the parameters given,

$$V_m = 120\sqrt{2} = 169.7\ \text{V}$$
$$Z = [R^2 + (\omega L)^2]^{0.5} = [2^2 + (377*0.02)^2]^{0.5} = 7.80\ \Omega$$
$$\theta = \tan^{-1}(\omega L/R) = \tan^{-1}(377*0.02/2) = 1.312\ \text{rad}$$
$$\omega\tau = \omega L/R = 377*0.02/2 = 3.77$$
$$\alpha = 45° = 0.785\ \text{rad}$$

(a) First, use Eq. (3-60) to determine if $\alpha = 45°$ is allowable. The minimum delay angle is:

$$\alpha_{\min} = \sin^{-1}\left(\frac{100}{120\sqrt{2}}\right) = 36°$$

Which indicates that 45° is allowable. Equation (3-61) becomes:

$$i(\omega t) = 21.8\sin(\omega t - 1.312) - 50 + 75.0e^{-\omega t/3.77} \text{ A} \quad \text{for } 0.785 \le \omega t \le 3.37 \text{ rad}$$

Where the extinction angle β is found numerically to be $3.37\ rad$ from the equation $i(\beta) = 0$.

(b) Power absorbed by the resistor is $I_{rms}^2 R$, where I_{rms} is computed from Eq. (3-59) using the preceding expression for $i(\omega t)$.

$$I_{\text{rms}} = \sqrt{\frac{1}{2\pi}\int_{\alpha}^{\beta} i^2(\omega t)\,d(\omega t)} = 3.90 \text{ A}$$

$$P = (3.90)^2(2) = 30.4 \text{ W}$$

(c) Power absorbed by the DC source is $I_o V_{dc}$, where I_o is computed from Eq. (3-58).

$$I_o = \frac{1}{2\pi}\int_{\alpha}^{\beta} i(\omega t)\,d(\omega t) = 2.19 \text{ A}$$

$$P_{\text{dc}} = I_o V_{\text{dc}} = (2.19)(100) = 219 \text{ W}$$

3.11 COMMUTATION

The Effect of Source Inductance

The preceding discussion on half-wave rectifiers assumed an ideal source. In practical circuits, the source has an equivalent impedance that is predominantly inductive reactance. For the single-diode half-wave rectifiers of Figs. 3-1 and 3-2, the nonideal circuit is analyzed by including the source inductance with the load elements.

However, the source inductance causes a fundamental change in circuit behavior for circuits like the half-wave rectifier with a freewheeling diode. A half-wave rectifier with a freewheeling diode and source inductance *Ls* is shown in Fig. 3-18*a*.

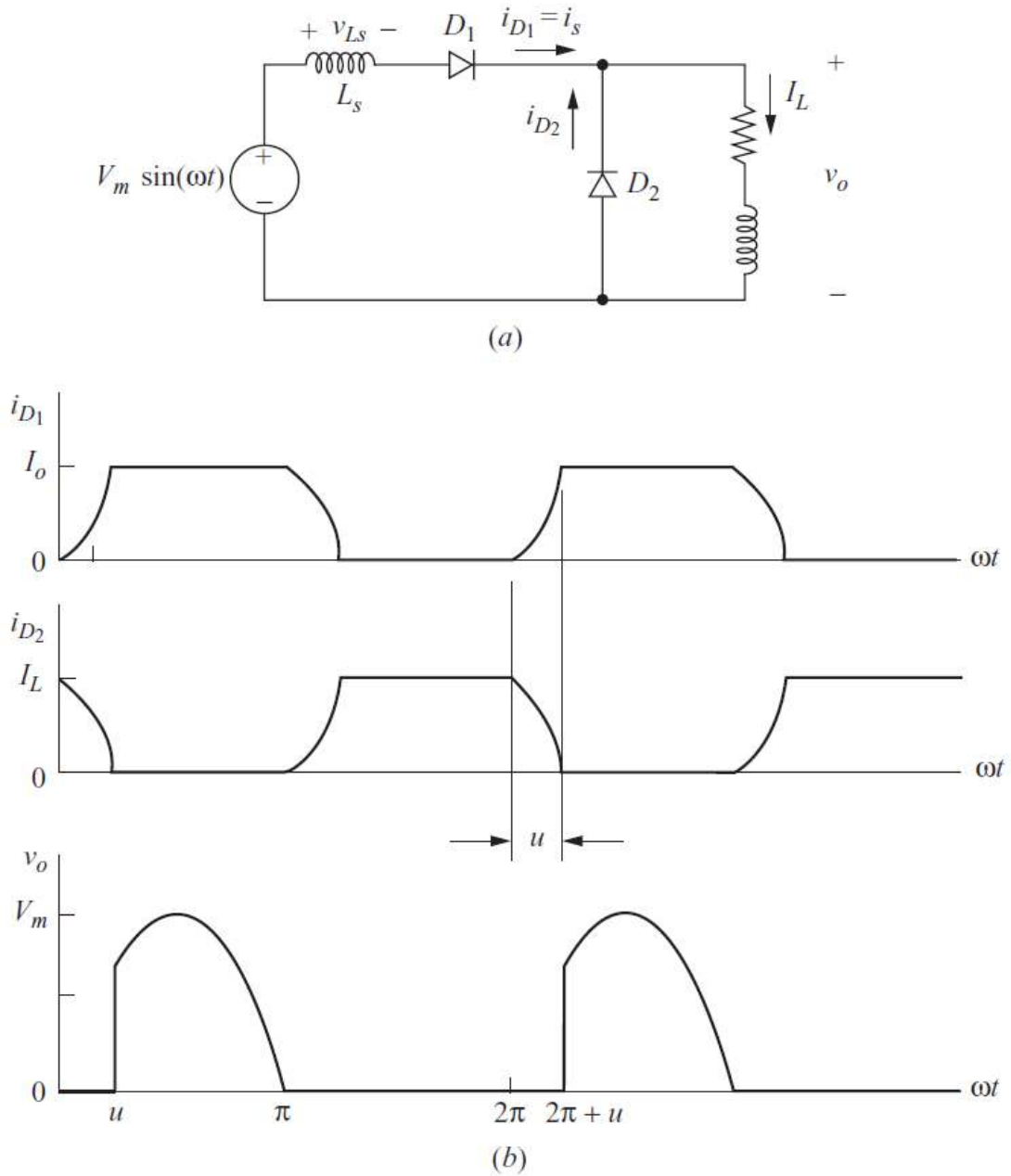


Figure 3-18 (*a*) Half-wave rectifier with freewheeling diode and source inductance; (*b*) Diode currents and load voltage showing the effects of Commutation.

Assume that the load inductance is very large, making the load current constant. At $t = 0^-$, the load current is I_L, D_1 is off, and D_2 is on.
As the source voltage becomes positive, D_1 turns on, but the source current does not instantly equal the load current because of L_S. Consequently, D_2 must remain on while the current in L_S and D_1 increases to that of the load.

The interval when both D_1 and D_2 are on is called the commutation time or commutation angle.

Commutation is the process of turning off an electronic switch, which usually involves transferring the load current from one switch to another. Commutation in this case is an example of natural commutation or line commutation, where the change in instantaneous line voltage results in a device turning off. Other applications may use forced commutation, where current in a device such as a thyristor is forced to zero by additional circuitry. Load commutation makes use of inherent oscillating currents produced by the load to turn a device off.

When both D_1 and D_2 are on, the voltage across L_S is

$$v_{Ls} = V_m \sin(\omega t) \tag{3-62}$$

And current in L_S and the source is:

$$i_s = \frac{1}{\omega L_s}\int_0^{\omega t} v_{Ls}\, d(\omega t) + i_s(0) = \frac{1}{\omega L_s}\int_0^{\omega t} V_m \sin(\omega t)\, d(\omega t) + 0$$

$$i_s = \frac{V_m}{\omega L_s}(1 - \cos \omega t) \tag{3-63}$$

Current in D_2 is

$$i_{D_2} = I_L - i_s = I_L - \frac{V_m}{\omega L_s}(1 - \cos \omega t)$$

The current in D_2 starts at I_L and decreases to zero. Letting the angle at which the current reaches zero be $\omega t = u$,

$$i_{D_2}(u) = I_L - \frac{V_m}{\omega L_s}(1 - \cos u) = 0$$

Solving for u,

$$u = \cos^{-1}\left(1 - \frac{I_L \omega L_s}{V_m}\right) = \cos^{-1}\left(1 - \frac{I_L X_s}{V_m}\right) \tag{3-64}$$

where $X_S = \omega L_S$ is the reactance of the source. Figure 3-18b shows the effect of the source reactance on the diode currents. The commutation from D_1 to D_2 is analyzed similarly, yielding an identical result for the commutation angle u.

The commutation angle affects the voltage across the load. Since the voltage across the load is zero when D_2 is conducting, the load voltage remains at zero through the commutation angle, as shown in Fig. 3-17b. Recall that the load voltage is a half-wave rectified sinusoid when the source is ideal. Average load voltage is:

$$V_o = \frac{1}{2\pi}\int_u^{\pi} V_m \sin(\omega t)\, d(\omega t)$$

$$= \frac{V_m}{2\pi}[-\cos(\omega t)]\Big|_u^{\pi} = \frac{V_m}{2\pi}(1 + \cos u)$$

Using u from Eq. (3-64),

$$\boxed{V_o = \frac{V_m}{\pi}\left(1 - \frac{I_L X_s}{2V_m}\right)} \qquad (3\text{-}65)$$

Recall that the average of a half-wave rectified sine wave is $\frac{V_m}{\pi}$. Source reactance thus reduces average load voltage.

3.12 Summary

• A rectifier converts ac to dc. Power transfer is from the ac source to the DC load.

• The half-wave rectifier with a resistive load has an average load voltage of $\frac{V_m}{\pi}$ and an average load current of $\frac{V_m}{\pi R}$.

• The current in a half-wave rectifier with an *RL* load contains a natural and a forced response, resulting in:

$$i(\omega t) = \begin{cases} \frac{V_m}{Z}\left[\sin(\omega t - \theta) + \sin(\theta)e^{-\omega t/\omega\tau}\right] & \text{for } 0 \le \omega t \le \beta \\ 0 & \text{for } \beta \le \omega t \le 2\pi \end{cases}$$

where $Z = \sqrt{R^2 + (\omega L)^2}$, $\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$ and $\tau = \frac{L}{R}$

The diode remains on as long as the current is positive. Power in the RL load is $I_{rms}^2\, R$.

• A half-wave rectifier with an RL-source load does not begin to conduct until the AC source reaches the DC voltage in the load. Power in the resistance is $I_{rms}^2\, R$, and power absorbed by the dc source is $I_o\, V_{dc}$, where I_o is the average load current.

The load current is expressed as:

$$i(\omega t) = \begin{cases} \frac{V_m}{Z}\sin(\omega t - \theta) - \frac{V_{dc}}{R} + Ae^{-\omega t/\omega\tau} & \text{for } \alpha \le \omega t \le \beta \\ 0 & \text{otherwise} \end{cases}$$

Where

$$A = \left[-\frac{V_m}{Z}\sin(\alpha - \beta) + \frac{V_{\text{dc}}}{R}\right]e^{\alpha/\omega\tau}$$

A freewheeling diode forces the voltage across an RL load to be a half-wave rectified sine wave. The load current can be analyzed using Fourier analysis. A large load inductance results in a nearly constant load current.

• A large filter capacitor across a resistive load makes the load voltage nearly constant. Average diode current must be the same as average load current, making the peak diode current large.

• An SCR in place of the diode in a half-wave rectifier provides a means of controlling output current and voltage.